

GaussianMarkovRandomFields.jl: Flexible Latent Gaussian Modeling with Sparse Precision in Julia

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Summary

Bayesian inference provides a strong theoretical foundation to draw conclusions from real-world data, with applications across all of the sciences. Yet in practice, algorithms for Bayesian inference often require excessive computational resources, limiting their applicability to large-scale problems. Particularly in the context of spatial and spatiotemporal settings, Gaussian Markov Random Fields (GMRFs) ([Håvard Rue & Held, 2005](#)) offer a remedy to this issue. GMRFs are Gaussian distributions with sparse precision matrices. Inference under these models scales particularly favorably due to the use of sparse linear algebra routines. This computational efficiency has led to their widespread adoption across various statistical applications ([Lindgren et al., 2022](#)).

GaussianMarkovRandomFields.jl provides a flexible and efficient Julia implementation of GMRFs. The package includes various methods to construct GMRFs, including the stochastic partial differential equation (SPDE) approach ([Lindgren et al., 2011](#)), and provides efficient, customizable routines for GMRF computations. It is designed to be intuitive to use for rapid prototyping, yet sufficiently flexible to empower expert users to solve advanced problems. As such, it is suitable for both research and teaching in spatial statistics and Bayesian modeling.

Statement of Need

GMRFs have become a cornerstone of modern spatial statistics, yet practical barriers limit their accessibility. Implementing GMRF-based models requires navigating sparse linear algebra libraries, finite element discretizations, and problem-specific solver selection, all of which create a steep barrier to entry for domain scientists.

GaussianMarkovRandomFields.jl eliminates these barriers by providing a feature-rich toolkit for creation and manipulation of GMRFs. The package offers utilities to create well-known latent models, ranging from simple autoregressive and Besag ([Besag & Kooperberg, 1995](#)) models to more involved spatial and spatiotemporal finite element-based models ([Clarotto et al., 2024](#); [Lindgren et al., 2011](#)). For users requiring continuous spatial domains, the SPDE approach is implemented through integration with Ferrite.jl ([Carlsson et al., 2025](#)), enabling principled approximations of Matérn and related Gaussian processes on arbitrary geometries. As an alternative to SPDE-based constructions, the package also implements KL-minimizing sparse Cholesky approximations ([Schäfer et al., 2021](#)), which construct sparse precision matrices directly from covariance specifications and are particularly effective for covariance functions with strong screening effects.

The package extends beyond Gaussian observations through Gaussian approximation methods. When working with non-Gaussian data such as counts or binary outcomes, the package seamlessly handles exponential family likelihoods. Users may also specify custom likelihood functions, for which the package exploits sparsity structure through automatic differentiation

to maintain computational efficiency even for complex observation models, an approach established by Skaug and Fournier (Skaug & Fournier, 2006) and popularised by TMB (Kristensen et al., 2016). For large-scale applications, the package provides efficient marginal variance computation methods (Lin et al., 2011; H. Rue, 2005; Sidén et al., 2018). Figure 1 demonstrates this capability on a spatial binary classification task. Automatic differentiation support via ForwardDiff.jl (Revels et al., 2016), Enzyme.jl (Moses & Churavy, 2020), Zygote.jl (Innes, 2018), Mooncake.jl (Tebbutt & Ge, 2024) and DifferentiationInterface.jl (Dalle & Hill, 2025) enable gradient-based optimization and inference methods, making the package compatible with modern probabilistic programming frameworks including Turing.jl (Ge et al., 2018).

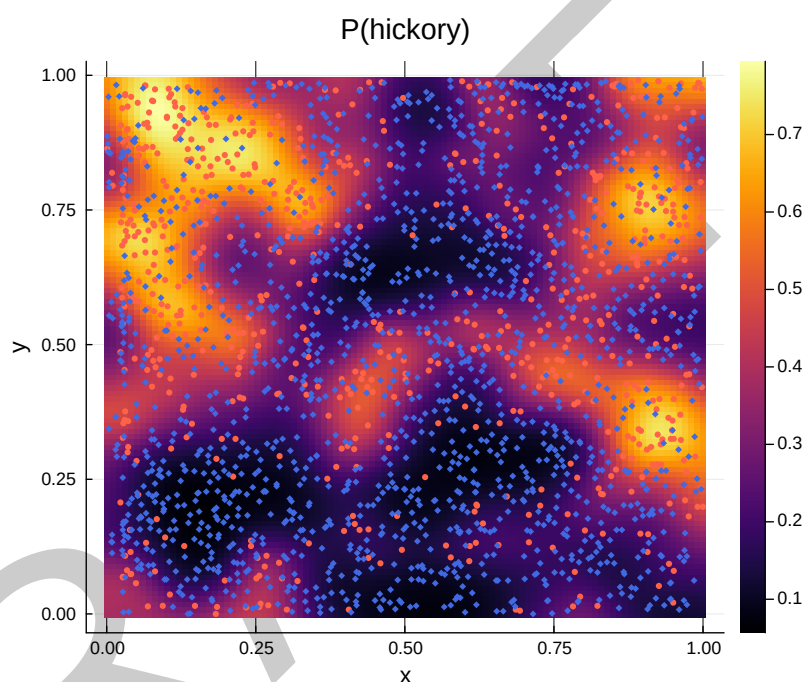


Figure 1: Spatial binary classification of tree species in the Lansing Woods dataset (Baddeley et al., 2015; Gerrard, 1969) using a Matérn latent field with Bernoulli observations. Points show observed trees: hickory (circles) and other species (diamonds). The heatmap shows predicted probabilities, demonstrating the package's support for non-Gaussian likelihoods through efficient Gaussian approximation.

State of the Field

Implementations of GMRF-based inference have historically centered on R. R-INLA (Håvard Rue et al., 2009) provides comprehensive support for latent Gaussian models through integrated nested Laplace approximations, but its model library is curated, and custom extensions require low-level C implementation. The inlabru package (Bachl et al., 2019) adds a flexible modeling interface for non-linear predictors while remaining tied to R-INLA's computational backend. The rSPDE package (Bolin & Simas, 2025) supports fractional SPDE models with non-integer smoothness. Template Model Builder (TMB) (Kristensen et al., 2016) supports Laplace approximation and gradient-based inference over latent Gaussian models via C++ templates, but requires users to write C++ for their models. In Python, general-purpose probabilistic programming frameworks such as PyMC (Abril-Pla et al., 2023) expose some GMRF primitives but lack specialized machinery.

Across these packages, GMRFs appear as implementation details of a particular inference algorithm, buried in C or C++ code and not exposed as first-class objects that users can

65 extend. `GaussianMarkovRandomFields.jl` instead treats the GMRF as the primary abstraction,
 66 exposed through a high-level interface, which enables workflows that are impractical elsewhere.

67 Neither the observation likelihood nor the latent prior need come from a fixed catalogue: both
 68 may be ordinary Julia functions, and the sparsity of their Hessians is discovered automatically.
 69 A latent field whose own dynamics are nonlinear (e.g., a state-space model of the kind routinely
 70 written in C++ for TMB (Nielsen & Berg, 2014)) is therefore expressed directly, and the
 71 iterated Laplace approximation such a model requires follows from the same machinery. Because
 72 a GMRF is itself a distribution with automatic differentiation rules attached, it composes
 73 directly with probabilistic programming languages: gradient-based samplers such as NUTS
 74 (Hoffman & Gelman, 2014) target the latent field and its hyperparameters jointly, giving fully
 75 Bayesian inference (Ge et al., 2018). SPDE discretizations are delegated to a general finite
 76 element library (Carlsson et al., 2025), putting custom differential operators, meshes, and
 77 boundary conditions within reach; the capability underlying (Weiland et al., 2025).

78 Software Design

79 The package is organized around an abstract type hierarchy centered on `AbstractGMRF`.
 80 Concrete subtypes, including the default GMRF, the hard-constrained `ConstrainedGMRF`, and the
 81 model-wrapping `MetaGMRF`, expose a uniform interface for mean, precision, `logpdf`, sampling,
 82 and marginal variance, so downstream code is written once and works across variants.

83 A separate `LatentModel` abstraction decouples model specification (hyperparameters mapped to
 84 sparse precision) from construction of GMRF instances. Provided models include autoregressive
 85 processes $AR(p)$, random walks, IID, fixed effects, Besag, BYM2, Matérn SPDE, and
 86 compositions thereof; users add their own by implementing a small protocol. A formula
 87 interface built on `StatsModels.jl` offers the familiar R-style syntax for combined models.

88 Linear solves are delegated to `LinearSolve.jl`, so users can swap `CHOLMOD`, `Pardiso`,
 89 `LDLT` factorizations, or preconditioned conjugate gradient without touching model code.
 90 Automatic differentiation is provided through package extensions for `ForwardDiff.jl`,
 91 `Zygote.jl/ChainRulesCore.jl`, `Enzyme.jl` and `Mooncake.jl`, with chain rules written at the
 92 GMRF-construction level so gradients flow through composed models. This split (abstract
 93 core, composable solver backends, AD via extensions) keeps the package's load-time cost
 94 small while preserving extensibility.

95 Research Impact Statement

96 The package has supported methodological research on probabilistic PDE solvers (Weiland et al.,
 97 2025). Near-term impact is rooted in the package's scope: `GaussianMarkovRandomFields.jl`
 98 supplies the foundational infrastructure needed to bring well-established latent-Gaussian
 99 inference methodologies to the Julia ecosystem.

100 Integrated Nested Laplace Approximation (INLA) (Håvard Rue et al., 2009) and Template Model
 101 Builder (TMB) (Kristensen et al., 2016), with HMC extensions such as `tmbstan` (Monnahan
 102 & Kristensen, 2018), are widely used in ecology, epidemiology, and environmental statistics,
 103 but have been available only in R or as C++ templates. The sparse-linear-algebra, automatic-
 104 differentiation, and latent-model abstractions provided here are the prerequisite machinery for
 105 Julia implementations of these methods. `Latte.jl` (Weiland & Latte.jl contributors, 2026), a
 106 probabilistic programming language for latent Gaussian models built by the author on these
 107 abstractions, provides INLA, TMB-style Laplace approximation, and HMC-Laplace behind a
 108 model-specification syntax — evidence that the machinery described here is sufficient for that
 109 purpose, not merely intended for it. Integration with `Turing.jl` (Ge et al., 2018) additionally
 110 enables full MCMC over GMRF hyperparameters, combining GMRF sub-models with arbitrary
 111 non-Gaussian components in a probabilistic program.

112 The package targets researchers and practitioners working in spatial statistics, epidemiology,
 113 environmental science, and related fields, and can serve both as a turnkey toolkit and as a
 114 substrate for methodological development in Julia's AD- and SciML-aware ecosystem.

115 AI Usage Disclosure

116 Claude Code (Anthropic's CLI coding assistant) was used as a development aid during the late
 117 stages of implementation, testing, and documentation, and when drafting portions of this paper.
 118 All design decisions, mathematical specifications, and package scope are the author's own.
 119 AI assistance was used to accelerate mechanical tasks (code translation, docstring drafting,
 120 routine refactors, prose tightening) under the author's direction. Every AI-produced suggestion
 121 was reviewed, edited, and integrated by the author before commit or inclusion. Correctness
 122 was verified through the package's test suite, which cross-checks implementations against
 123 dense-matrix baselines, finite-difference gradients, and analytical solutions where available, and
 124 by running the accompanying documentation examples. No other generative-AI tools were
 125 used.

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